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TITLE PAGE

- Problem Statement ID – 26202
- Problem Statement Title- NET COMPASS: Intelligent Cellular Network Coverage & Performance Monitoring System
- Theme-
- PS Category- Software
- Team NET COMPASS — Dept. of Electronics & Communication Engineering

IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- NET COMPASS identifies the available cellular network type (5G, 4G, LTE or H+) at the user’s current location, using an ESP32 controller paired with a SIM7600 4G/LTE modem to measure signal strength, download/upload speed and latency in real time.
- Users can instantly see whether they are in a strong or weak coverage zone and pick a better nearby location for reliable connectivity, instead of guessing why their network is poor.
- Combines live signal analytics (RSSI, RSRP, RSRQ, SINR) with on-device OLED display and SD-card logging in one low-cost, portable unit — a foundation for future GPS-based coverage mapping and AI-driven network analysis.

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TECHNICAL APPROACH

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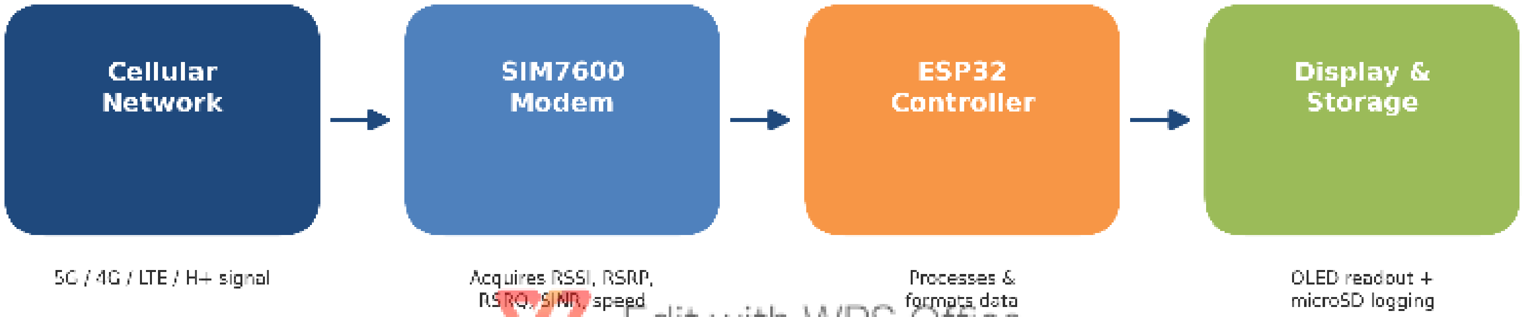
- Technologies used: ESP32 DevKit (main controller), SIM7600 4G/LTE modem, 0.96” OLED display, microSD module for data logging; firmware written in C/C++ (Arduino framework) with AT-command based modem communication.
- Signal flow: Cellular Network → SIM7600 Modem → ESP32 Controller → Display & Storage. Demonstration workflow: Power ON → Initialize SIM7600 → Register with Network → Acquire Parameters (RSSI/RSRP/RSRQ/SINR, speed, latency) → Display on OLED → Log to SD Card → Repeat.



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FEASIBILITY AND VIABILITY

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- Feasibility: built from off-the-shelf, low-cost components (ESP32, SIM7600, OLED, microSD) with a proven signal-flow architecture; Semester 1 budget of ₹4,000–₹7,000 keeps the prototype easily buildable and testable.
- Challenges & risks: accuracy and stability of live modem readings, a deliberately limited scope (no GPS, cloud or ML in this phase), and dependency on SIM7600 network registration in low-coverage test areas.
- Strategies: keep Semester 1 focused strictly on accurate, stable core data acquisition before scaling; phase GPS integration, cloud coverage mapping and AI-based analysis into Years 2 & 3 to manage complexity incrementally.

IMPACT AND BENEFITS

- Impact: benefits rural connectivity assessment, embedded-systems education, field engineer diagnostics and telecom network research by making invisible network quality visible and measurable at any location.
- Benefits: Social — helps underserved/rural users find reliable connectivity; Economic — low-cost (₹4,000–₹7,000) alternative to expensive diagnostic equipment; Educational — hands-on embedded/telecom learning platform; scalable foundation for future coverage-mapping and AI-based services.

Rural Connectivity

Economic Impact

Embedded Education

Telecom Research

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RESEARCH AND REFERENCES

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- Key network parameters monitored follow standard 3GPP-defined cellular metrics: RSSI (Received Signal Strength Indicator), RSRP (Reference Signal Received Power), RSRQ (Reference Signal Received Quality) and SINR (Signal-to-Interference-plus-Noise Ratio).
- Technical references: SIMCom SIM7600 series AT command manual & hardware design guide; Espressif ESP32 technical reference manual.
- Prototype architecture and roadmap benchmarked against standard 4G/LTE network diagnostic practices used in field engineering and telecom research.